

Hardness → Hardness is due to the presence of calcium & magnesium dissolved salt & heavy metals in water.

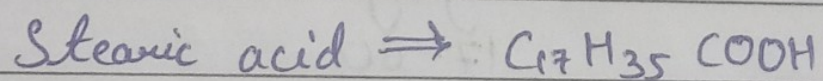
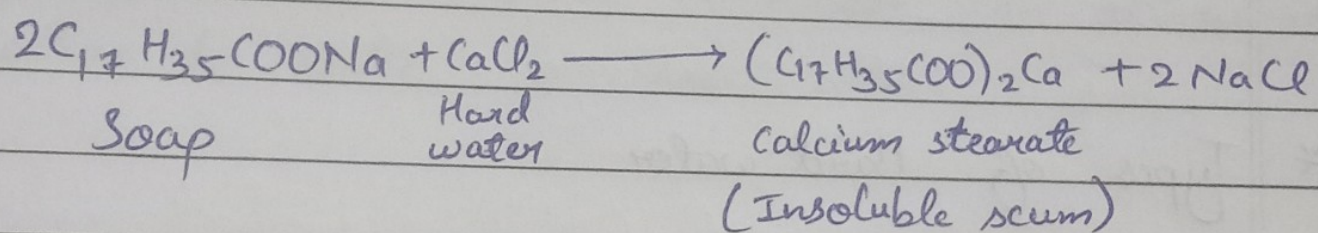
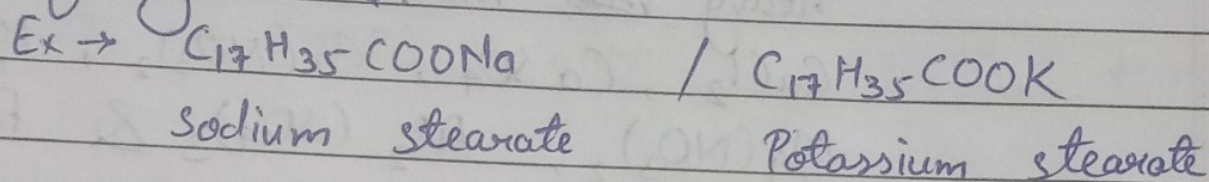
OR

→ Soap consuming capacity of water.

OR

→ Hardness prevents lathering of soap.

Soap → Sodium (Na) or Potassium (K) salts of higher fatty acids.



Classification of water

Water is of 2 types -

1) Soft water → Water which produces lather with soap.  
Ex →  $C_{17}H_{35}COONa + H_2O \longrightarrow C_{17}H_{35}COOH + NaOH$   
Stearic acid

2) Hard water → Water which do not form lather with soap.  
Ex →  $C_{17}H_{35}COONa + CaCl_2 \longrightarrow (C_{17}H_{35}COO)_2Ca + NaCl$   
Insoluble scum

\* Types of Hardness →  
It is of two types

1> Temporary Hardness → It is caused due to the presence of  $\text{Ca}^{++}/\text{Mg}^{++}$  bicarbonate ( $\text{HCO}_3^-$ ) i.e.,  $\text{Ca}(\text{HCO}_3)_2 / \text{Mg}(\text{HCO}_3)_2$

2> Permanent Hardness → It is caused due to the presence of chlorides ( $\text{Cl}^-$ ), sulphate ( $\text{SO}_4^{--}$ ) or nitrates ( $\text{NO}_3^-$ ) of  $\text{Ca}^{++}/\text{Mg}^{++}$  & salts of some heavy ~~met~~ metals i.e.,  $\text{CaCl}_2$ ,  $\text{CaSO}_4$ ,  $\text{Ca}(\text{NO}_3)_2$ ,  $\text{MgCl}_2$ ,  $\text{MgSO}_4$ ,  $\text{Mg}(\text{NO}_3)_2$ ,  $\text{Al}_2(\text{SO}_4)_3$  &  $\text{FeSO}_4$ .

\* Types of Hard water

1> Soft water  
Hardness 0-17  
(ppm)

2> Slightly H.W.  
17-60

3> Moderately H.W.  
60-120

(B)

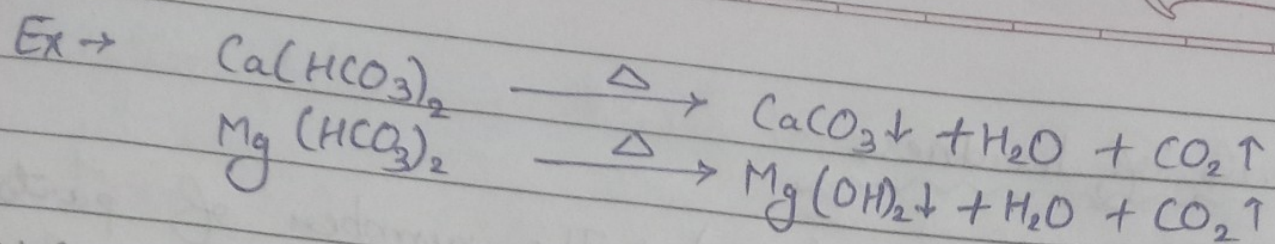
4> Hard water  
120-180

5> Very H.W.  
Greater than 180

\* Removal of Hard water

(A) Temporary Hardness → By 2 methods

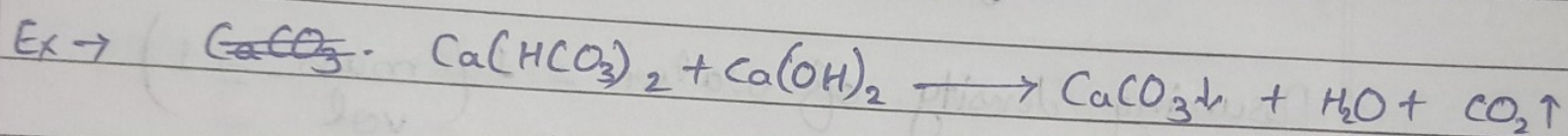
(a) Boiling ⇒ Temp. Hardness can be removed by boiling.



Note →  $\text{MgCO}_3$  is not precipitate, it is soluble.

(b) Clark's method → Clark reagent is  $\text{Ca(OH)}_2$  (lime).

$\text{CaCO}_3$  &  $\text{Mg(OH)}_2$  are the precipitates which can be removed by filtration & soft water is obtained.



The precipitate of Calcium carbonate can be removed by filtration & water is made soft.

(B) Permanent Hardness → By 3 methods

1) Lime soda Process → upto 50 ppm

2) Zeolite Process → upto 10 ppm

3) Ion Exchange Process → upto 2 ppm

## Units of hardness

1) Parts per million (ppm)  $\rightarrow$  The number of parts of  $\text{CaCO}_3$  equivalent hardness present in  $10^6$  parts of water.

2) mg/lit  $\rightarrow$  The number of <sup>mg</sup> parts of  $\text{CaCO}_3$  equivalent hardness present in 1 lit of water.

$$\underline{1 \text{ mg/lit} = 1 \text{ ppm}}$$

Assuming density of water = 1 ( $\frac{\text{mass}}{\text{vol}} = 1$ )

$$1 \text{ kg} = 1 \text{ lit}$$

$$1 \text{ kg} = 10^3 \text{ g} = 10^6 \text{ mg} = 1 \text{ lit}$$

mg/lit  $\Rightarrow$  1 mg of  $\text{CaCO}_3$  equivalent hardness present in 1 lit or  $10^6$  mg of water.

$\Rightarrow$  1 part of  $\text{CaCO}_3$  equivalent hardness present in  $10^6$  parts of water = 1 ppm.

Therefore,  $1 \text{ mg} = 1 \text{ ppm}$

3) Degree Clark ( $^\circ\text{Cl}$ )  $\Rightarrow$  The number of parts of  $\text{CaCO}_3$  equivalent hardness present in 70,000 parts of water.

4) Degree French ( $^\circ\text{Fr}$ )  $\Rightarrow$  The number of parts of  $\text{CaCO}_3$  equivalent hardness present in  $10^5$  parts of water.

\* Relation b/w units of hardness

$$1 \text{ ppm} = 1 \text{ mg/lit} = 0.07^\circ \text{Cl} = 0.1^\circ \text{Fr}$$

\* Classification of water

\* Degree of Hardness

⇒ The amount of substance of  $\text{Ca}^{++}$  and  $\text{Mg}^{++}$  present in certain volume of water known as degree of hardness

# Expressing hardness in terms of  $\text{CaCO}_3$  equivalent = ?

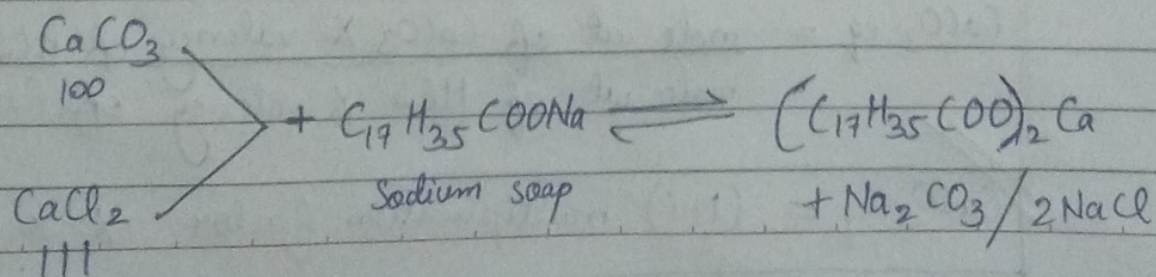
Because ⇒ It is most insoluble salt which can removed by filtration.

2) Its molecular weight is exactly 100 which makes calculation easy.

Conversion of hardness in  $\text{CaCO}_3$  equivalent

Principle → Soap consuming capacity of water is known as its hardness.

If two salts consumes same amount of soap means both shows same value of hardness.



111 parts of by weight of  $\text{CaCl}_2$  reacts with same amount of salt as 100 parts by weight of  $\text{CaCO}_3$  reacts.

$$111 \text{ parts of } \text{CaCl}_2 = 100 \text{ parts of } \text{CaCO}_3$$

$$1 \text{ part of } \text{CaCl}_2 = \frac{100}{111} \text{ parts of } \text{CaCO}_3$$

$$x \text{ parts of } \text{CaCl}_2 = x \times \frac{100}{111} \text{ parts of } \text{CaCO}_3$$

General formula  $\rightarrow \frac{\text{mole wt of } \text{CaCO}_3}{\text{mole wt of Hardness causing salt (HCS)}} \times x$   
of  $\text{CaCO}_3$  eq.

$x$  = mass or strength of HCS in water

Amount of soap consumed by any salt depends on its valency, therefore the formula of  $\text{CaCO}_3$  equivalents will be

$$= \frac{\frac{\text{mole wt of } \text{CaCO}_3}{\text{valency of } \text{CaCO}_3}}{\frac{\text{mole wt of HCS}}{\text{valency of HCS}}} \times x$$

$$\text{CaCO}_3 \text{ eq} = \frac{\text{mole wt of } \text{CaCO}_3 \times \text{valency of HCS}}{\text{mole wt of HCS} \times \text{valency of } \text{CaCO}_3} \times x$$

$$\text{CaCO}_3 \text{ eq} = (\text{M.F.}) \text{ multiplication factor } \times x$$

\* Formula for

$\text{CaCO}_3$  eq.

$\text{CaCO}_3$  eq.

| S.No. | Ca  |
|-------|-----|
| 1     | C   |
| 2     | M   |
| 3     | C   |
| 4     |     |
| 5     |     |
| 6     |     |
| 7     |     |
| 8     |     |
| 9     | C   |
| 10    |     |
| 11    | Fes |

Some atoms

Ca = 40

Mg = 24

S = 32

Na = 23

Al = 27

Cl<sub>2</sub> reacts with  
parts by weight

of CaCO<sub>3</sub>

of CaCO<sub>3</sub>

parts of CaCO<sub>3</sub>

CaCO<sub>3</sub> x x  
ess  
(S)

water

salt depends  
formula of CaCO<sub>3</sub>

of HCl x x  
of CaCO<sub>3</sub>

x

\* Formula for conversion of Divalent HCS into CaCO<sub>3</sub> eq. (Ca<sup>++</sup>/Mg<sup>++</sup>)

$$\text{CaCO}_3 \text{ eq.} = \frac{\text{Mol. wt. of CaCO}_3}{\frac{\text{Mol. wt. of HCS}}{2}} \times \text{Mass of HCS}$$

$$\text{CaCO}_3 \text{ eq.} = \frac{\text{Mol. wt. of CaCO}_3}{\text{Mol. wt. of HCS}} \times \text{Mass of HCS}$$

| S.No. | Constituents                          | Mol. wt. | Multiplication factor |
|-------|---------------------------------------|----------|-----------------------|
| 1     | Ca (HCO <sub>3</sub> ) <sub>2</sub>   | 162      | 100/162               |
| 2     | Mg (HCO <sub>3</sub> ) <sub>2</sub>   | 146      | 100/146               |
| 3     | Ca SO <sub>4</sub>                    | 136      | 100/136               |
| 4     | Mg SO <sub>4</sub>                    | 120      | 100/120               |
| 5     | CaCl <sub>2</sub>                     | 111      | 100/111               |
| 6     | Mg Cl <sub>2</sub>                    | 95       | 100/95                |
| 7     | Ca (NO <sub>3</sub> ) <sub>2</sub>    | 164      | 100/164               |
| 8     | Mg (NO <sub>3</sub> ) <sub>2</sub>    | 148      | 100/148               |
| 9     | Ca CO <sub>3</sub>                    | 100      | 100/100               |
| 10    | Mg CO <sub>3</sub>                    | 84       | 100/84                |
| 11    | FeSO <sub>4</sub> . 7H <sub>2</sub> O | 278      | 100/278               |

Some atomic weights

Ca = 40

Mg = 24

S = 32

Na = 23

Al = 27

\* Formula for conversion of Monovalent HCS into  $\text{CaCO}_3$  eq.

$$\text{CaCO}_3 \text{ eq} = \frac{\text{Mol. wt of CaCO}_3}{\text{Mol. wt of HCS}} \times \text{Mass of HCS}$$

$$\text{CaCO}_3 \text{ eq} = \frac{\text{Mol. wt of CaCO}_3}{2 \times \text{Mol. wt of HCS}} \times \text{Mass of HCS}$$

| S.No. | Constituents            | Mol. wt. | Multiplication factor |
|-------|-------------------------|----------|-----------------------|
| 1     | $\text{H}^+$            | 1        | $100 / 2 \times 1$    |
| 2     | $\text{OH}^-$           | 17       | $100 / 2 \times 17$   |
| 3     | $\text{HCl}$            | 36.5     | $100 / 2 \times 36.5$ |
| 4     | $\text{H}_2\text{SO}_4$ | 98       | $100 / 2 \times 98$   |
| 5     | $\text{NaAlO}_2$        | 82       | $100 / 2 \times 82$   |
| 6     | $\text{HCO}_3^-$        | 61       | $100 / 2 \times 61$   |

\* Formula for conversion of Trivalent HCS into  $\text{CaCO}_3$  eq.

$$\text{CaCO}_3 \text{ eq.} = \frac{\text{Mol. wt of CaCO}_3}{\text{Mol. wt of HCS}} \times \text{Mass of HCS}$$

$$\text{CaCO}_3 \text{ eq.} = \frac{3 \times \text{Mol. wt of CaCO}_3}{2 \times \text{Mol. wt of HCS}} \times \text{Mass of HCS}$$

→ Formula for

$\text{CaCO}_3$

$\text{CaCO}_3$

| S.No. | Con           |
|-------|---------------|
| 1     | $\text{Al}_2$ |

Que → A sample of salt in mg/  
 $\text{Mg}(\text{HCO}_3)_2$  →  
 $\text{MgSO}_4$  → 40  
Calculate the

Sol →

HCS

$\text{Mg}(\text{HCO}_3)_2$

$\text{CaCl}_2$

$\text{Ca}(\text{HCO}_3)_2$

$\text{MgSO}_4$

$\text{MgCl}_2$

→ Formula for conversion of  $Al_2(SO_4)_3$  into  $CaCO_3$  eq.

$$CaCO_3 = \frac{\text{Mol wt of } CaCO_3}{2} \times \frac{\text{Mass of HCS } Al_2(SO_4)_3}{\text{Mol wt of HCS } Al_2(SO_4)_3}$$

$$CaCO_3 = \frac{3 \times \text{Mol wt. of } CaCO_3}{\text{Mol wt of HCS } Al_2(SO_4)_3} \times \text{Mass of HCS } Al_2(SO_4)_3$$

factor

| S.No. | Constituents   | Mol. wt. | Multiplication factor |
|-------|----------------|----------|-----------------------|
| 1     | $Al_2(SO_4)_3$ | 342      | $3 \times 100 / 342$  |

Que → A sample of water is found to contain following dissolved salt in mg/lit.

$Mg(HCO_3)_2 \rightarrow 73 \text{ ppm}$ ,  $CaCl_2 \rightarrow 111 \text{ ppm}$ ,  $Ca(HCO_3)_2 \rightarrow 81 \text{ ppm}$ ,

$MgSO_4 \rightarrow 40 \text{ ppm}$  and  $MgCl_2 \rightarrow 95 \text{ ppm}$ .

Calculate the temporary and permanent hardness of water.

Sol →

| HCS           | x       | M.F.    | $CaCO_3 \text{ eq} \pm M.F. \times x$                     |
|---------------|---------|---------|-----------------------------------------------------------|
| $Mg(HCO_3)_2$ | 73 ppm  | 100/146 | $\frac{100}{146} \times 73 \Rightarrow 50 \text{ ppm}$    |
| $CaCl_2$      | 111 ppm | 100/111 | $\frac{100}{111} \times 111 \Rightarrow 100 \text{ ppm}$  |
| $Ca(HCO_3)_2$ | 81 ppm  | 100/162 | $\frac{100}{162} \times 81 \Rightarrow 50 \text{ ppm}$    |
| $MgSO_4$      | 40 ppm  | 100/120 | $\frac{100}{120} \times 40 \Rightarrow 33.33 \text{ ppm}$ |
| $MgCl_2$      | 95 ppm  | 100/95  | $\frac{100}{95} \times 95 \Rightarrow 100 \text{ ppm}$    |

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Page \_\_\_\_\_

$$\begin{aligned}\text{Temporary Hardness} &= \text{Ca}(\text{HCO}_3)_2 + \text{Mg}(\text{HCO}_3)_2 \\ &= 50 + 50 \\ &= 100 \text{ ppm}\end{aligned}$$

$$\begin{aligned}\text{Permanent Hardness} &= \text{CaCl}_2 + \text{MgSO}_4 + \text{MgCl}_2 \\ &= 100 + 33.33 + 100 \\ &= 233.33 \text{ ppm}\end{aligned}$$

$$\text{Total Hardness} = 100 + 233.33 = 333.33 \text{ ppm}$$